

# UNIT 4: PAGE REPLACEMENT SOLVED NUMERICAL EXAMPLES

**Course Pattern:** TE IT - 2024 Pattern (SPPU)

**Academic Term:** Semester V (2026)

**Document Type:** Solved Numerical Guide

**Course Code:** PCC-301-ITT

## Problem 1: Page Replacement Trace (FIFO, OPT, LRU)

### INTUITION:

#### Algorithm Rules:

- **FIFO:** Replace the page that was loaded earliest into RAM (queue order).
- **Optimal (OPT):** Replace the page that will not be referenced for the longest time in the future.
- **LRU:** Replace the page that has not been used for the longest time in the past.
- **Hit Ratio Calculation:**  $\text{Hit Ratio} = \frac{\text{Hits}}{\text{Total References}}$
- **Fault Ratio Calculation:**  $\text{Fault Ratio} = \frac{\text{Faults}}{\text{Total References}}$

### Problem Statement

Consider the page reference string: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1

For a system with **3 allocated physical RAM frames** (initially empty), compute the number of page faults and page hits using:

- (a) FIFO Algorithm
- (b) Optimal Page Replacement (OPT) Algorithm
- (c) Least Recently Used (LRU) Algorithm

### Step-by-Step Solution

#### 1.2.1 (a) First-In, First-Out (FIFO) Algorithm (3 Frames)

Trace of 3-frame queue over time:

- 7: [7, -, -] (Fault 1)
- 0: [7, 0, -] (Fault 2)
- 1: [7, 0, 1] (Fault 3)
- 2: Replaces 7 to [2, 0, 1] (Fault 4)
- 0: [2, 0, 1] (Hit 1)
- 3: Replaces 0 to [2, 3, 1] (Fault 5)
- 0: Replaces 1 to [2, 3, 0] (Fault 6)
- 4: Replaces 2 to [4, 3, 0] (Fault 7)
- 2: Replaces 3 to [4, 2, 0] (Fault 8)
- 3: Replaces 0 to [4, 2, 3] (Fault 9)
- 0: Replaces 4 to [0, 2, 3] (Fault 10)
- 3: [0, 2, 3] (Hit 2)
- 2: [0, 2, 3] (Hit 3)
- 1: Replaces 2 to [0, 1, 3] (Fault 11)
- 2: Replaces 3 to [0, 1, 2] (Fault 12)
- 0: [0, 1, 2] (Hit 4)

- 1: [0, 1, 2] (Hit 5)
- 7: Replaces 0 to [7, 1, 2] (Fault 13)
- 0: Replaces 1 to [7, 0, 2] (Fault 14)
- 1: Replaces 2 to [7, 0, 1] (Fault 15)

**FIFO Results:**

- Total References: 20
- **Total Page Faults: 15**
- **Total Page Hits: 5**
- **Hit Ratio:**  $\frac{5}{20} = 25\%$

**1.2.2 (b) Optimal Page Replacement (OPT) Algorithm (3 Frames)**

- 7: [7, -, -] (Fault 1)
- 0: [7, 0, -] (Fault 2)
- 1: [7, 0, 1] (Fault 3)
- 2: 7 used farthest in future to Replaces 7 to [2, 0, 1] (Fault 4)
- 0: [2, 0, 1] (Hit 1)
- 3: 1 used farthest in future to Replaces 1 to [2, 0, 3] (Fault 5)
- 0: [2, 0, 3] (Hit 2)
- 4: 0 or 3 next; 2 used farthest to Replaces 2 to [4, 0, 3] (Fault 6)
- 2: 0 and 3 used next to Replaces 4 to [2, 0, 3] (Fault 7)
- 3: [2, 0, 3] (Hit 3)
- 0: [2, 0, 3] (Hit 4)
- 3: [2, 0, 3] (Hit 5)
- 2: [2, 0, 3] (Hit 6)
- 1: 2 and 0 used next to Replaces 3 to [2, 0, 1] (Fault 8)
- 2: [2, 0, 1] (Hit 7)
- 0: [2, 0, 1] (Hit 8)
- 1: [2, 0, 1] (Hit 9)
- 7: 2 not used again to Replaces 2 to [7, 0, 1] (Fault 9)
- 0: [7, 0, 1] (Hit 10)
- 1: [7, 0, 1] (Hit 11)

**Optimal Results:**

- **Total Page Faults: 9**
- **Total Page Hits: 11**
- **Hit Ratio:**  $\frac{11}{20} = 55\%$

**1.2.3 (c) Least Recently Used (LRU) Algorithm (3 Frames)**

- 7: [7, -, -] (Fault 1)
- 0: [7, 0, -] (Fault 2)
- 1: [7, 0, 1] (Fault 3)
- 2: Replaces 7 (least recent) to [2, 0, 1] (Fault 4)
- 0: [2, 0, 1] (Hit 1)
- 3: Replaces 1 to [2, 0, 3] (Fault 5)
- 0: [2, 0, 3] (Hit 2)
- 4: Replaces 2 to [4, 0, 3] (Fault 6)
- 2: Replaces 3 to [4, 0, 2] (Fault 7)
- 3: Replaces 0 to [4, 3, 2] (Fault 8)
- 0: Replaces 4 to [0, 3, 2] (Fault 9)
- 3: [0, 3, 2] (Hit 3)
- 2: [0, 3, 2] (Hit 4)
- 1: Replaces 0 to [1, 3, 2] (Fault 10)
- 2: [1, 3, 2] (Hit 5)
- 0: Replaces 3 to [1, 0, 2] (Fault 11)
- 1: [1, 0, 2] (Hit 6)

- 7: Replaces 2 to [1, 0, 7] (Fault 12)
- 0: [1, 0, 7] (Hit 7)
- 1: [1, 0, 7] (Hit 8)

**LRU Results:**

- **Total Page Faults: 12**
- **Total Page Hits: 8**
- **Hit Ratio:**  $\frac{8}{20} = 40\%$

## Problem 2: TLB Effective Access Time (EAT) Calculation

### Problem Statement

Consider a paging system with:

- Physical RAM access time  $m = 100$  ns.
- TLB lookup search time  $\varepsilon = 20$  ns.
- TLB Hit Ratio  $\alpha = 80\% = 0.80$ .

**Questions:**

- (a) Compute the Effective Access Time (EAT).
- (b) What will be the EAT if the TLB Hit Ratio increases to 98%?

### Step-by-Step Solution

**2.2.1 (a) EAT for  $\alpha = 80\%$** 

$$\text{EAT} = \alpha * (\varepsilon + m) + (1 - \alpha) * (\varepsilon + 2m)$$

- $\text{EAT} = 0.80 * (20 + 100) + (1 - 0.80) * (20 + 2 * 100)$
- $\text{EAT} = 0.80 * (120) + 0.20 * (220)$
- $\text{EAT} = 96 + 44 = 140$  ns

**2.2.2 (b) EAT for  $\alpha = 98\%$** 

- $\text{EAT} = 0.98 * (120) + 0.02 * (220)$
- $\text{EAT} = 117.6 + 4.4 = 122$  ns

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*End of Operating Systems Unit 4 Solved Examples — SPPU TE IT 2024 Pattern*